

Basic Matrix Operations

Adding, subtracting and scaling matrices, multiplying matrices including a real inventory model, and using determinants and matrix equations

The rules you need.

- **Add / subtract** only matrices of the same size, entry by corresponding entry.
- **Scalar multiple:** multiply *every* entry by the scalar.
- **Product:** the entry in row i , column j of AB is row i of A paired term by term with column j of B , then summed. AB exists only when the number of columns of A equals the number of rows of B .
- **Determinant** of a 2×2 matrix: the product of the main diagonal minus the product of the other diagonal.
- **Equal matrices** have equal corresponding entries — that is what turns a matrix equation into ordinary equations.

1. Let $A = \begin{bmatrix} 3 & -2 \\ 5 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 6 \\ 2 & -7 \end{bmatrix}$.

Find $A + B$.

Answer: _____

2. Let $A = \begin{bmatrix} 4 & -3 \\ 2 & 6 \end{bmatrix}$.

Find $3A$.

Answer: _____

3. Let $A = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 \\ -2 & 5 \end{bmatrix}$.

Find the product AB . Show the row-by-column pairing for at least one entry.

Answer: _____

4. Let $A = \begin{bmatrix} 5 & -2 & 7 \\ 1 & 4 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 & -1 \\ 6 & -4 & 5 \end{bmatrix}$.

Find $A - B$.

Answer: _____

5. Let $A = \begin{bmatrix} 1 & -2 \\ 3 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 4 \\ -1 & 2 \end{bmatrix}$.

Find $2A + 3B$.

Answer: _____

6. A chain has two stores. The stock matrix S lists how many units of three products each store holds, and the price column P lists the price of each product in dollars.

$$S = \begin{bmatrix} 12 & 8 & 5 \\ 9 & 14 & 6 \end{bmatrix} \quad P = \begin{bmatrix} 40 \\ 25 \\ 60 \end{bmatrix}$$

Row 1 of S is the north store and row 2 is the south store. Find the product SP . Each entry of the answer is one store's total stock value in dollars.

Answer: _____

7. Find the determinant of $M = \begin{bmatrix} 7 & 3 \\ 2 & 5 \end{bmatrix}$.

Answer: _____

8. Let $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ -1 & 3 \\ 4 & 2 \end{bmatrix}$.

Check that the product AB is defined, state its size, then compute it.

Answer: _____

9. Find the values of x and y that make these two matrices equal.

$$\begin{bmatrix} x+2 & 9 \\ 2 & y-1 \end{bmatrix} = \begin{bmatrix} 7 & 9 \\ 2 & 9 \end{bmatrix}$$

Answer: _____

10. A ticket seller sold adult tickets and child tickets. The situation gives the system

$$2x + 3y = 12, \quad x - y = 1.$$

Write this system as a single matrix equation of the form $AX = B$, where A is the coefficient matrix, X is the column of variables and B is the column of constants. Then find x and y .

Answer: _____